AIORI HACKATHON 3

A3-PS005-TC048

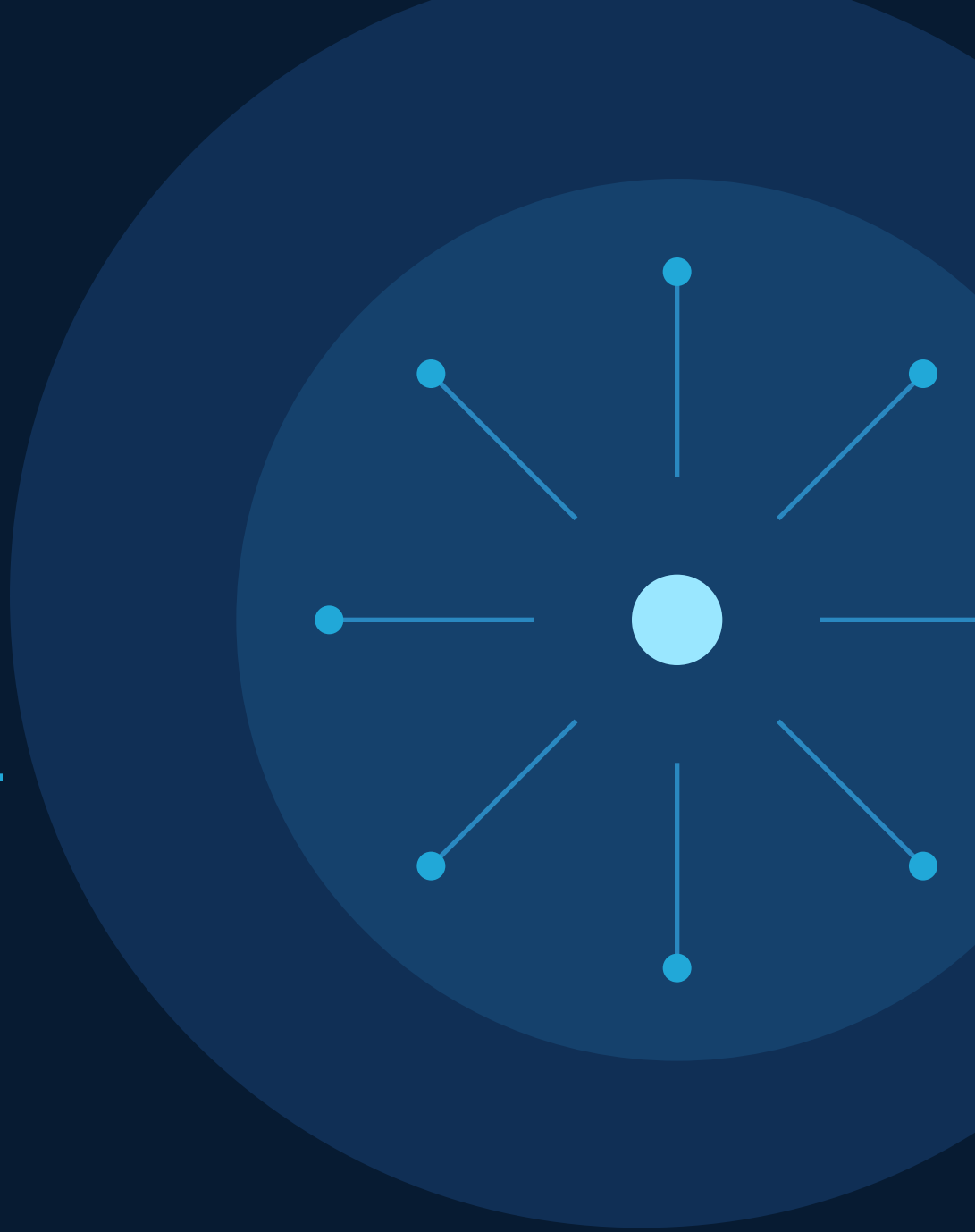
Explainable BGP Origin and Subprefix Anomaly Monitor

OriginGuard-X | Proposed Solution

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The problem we are solving

Detect suspicious routing changes without treating every unusual route as malicious.

Unexpected origin announcements

An AS can announce a prefix it does not own or is not authorized to originate. Misconfigurations and deliberate hijacks can look similar.

More-specific subprefix hijacks

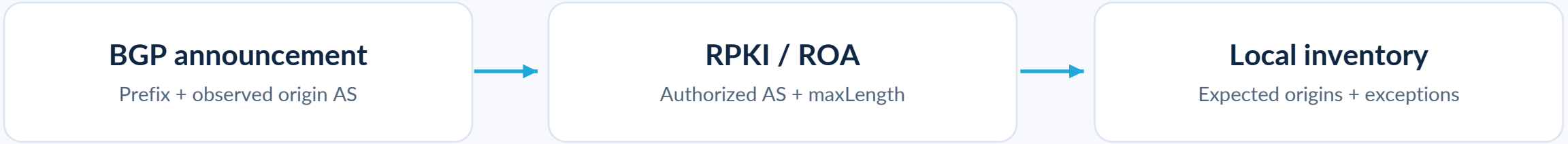
A /24 announced inside a legitimate /16 can attract traffic through longest-prefix matching, even if the aggregate remains visible.

WHAT PS-005 REQUIRES

- Replay BGP updates; correlate ROA/RPKI and operator prefix inventory.
- Detect origin and subprefix anomalies; explain each alert.
- Evaluate precision, recall, false positives, and alert latency.

Why origin validation alone is not enough

BGP announces reachability; RPKI adds authorization evidence, not proof of intent.



VALID INVALID NOTFOUND

ROA origin validation produces three states. Invalid signals a conflict with an applicable authorization; NotFound means no applicable authorization record. Neither state alone proves malicious intent. [1, 2]

Legitimate MOAS, origin migrations, delegated subprefixes, and maintenance windows must be represented as test controls - not silently treated as attacks.

What existing work tells us

Research-informed design; proposed novelty must be established through evaluation.

- **ARTEMIS - IEEE/ACM ToN (2018) [4]**
  Fast operator-centric BGP hijack detection; strong existing prior art.
- **BGPStream - ACM IMC (2016) [5]**
  Practical framework for replaying historical and live BGP observations.
- **RFC 6811 + RFC 8481 [1, 3]**
  Standard origin-validation semantics and operational clarifications.
- **RFC 9582 (2024) [2]**
  Current ROA object profile; replaces older RFC 6482.

Research angle to test: Can evidence correlation reduce false alerts vs. single-signal baselines?

OriginGuard-X: Evidence-first BGP monitoring

A transparent rule engine first; optional ML only after the baseline is measured.

01

Ingest

RIPE RIS / RouteViews MRT replay

02

Validate

ROA / VRP snapshot + maxLength

03

Correlate

Inventory, covering prefix, history

04

Explain

Triggered rules + counter-evidence

EXPLAINABLE ALERT

Observed AS64501 | 203.0.113.0/24 | RPKI Invalid

Evidence: unauthorized origin + more-specific route + inventory mismatch

Output: investigate; never auto-block or blackhole routes

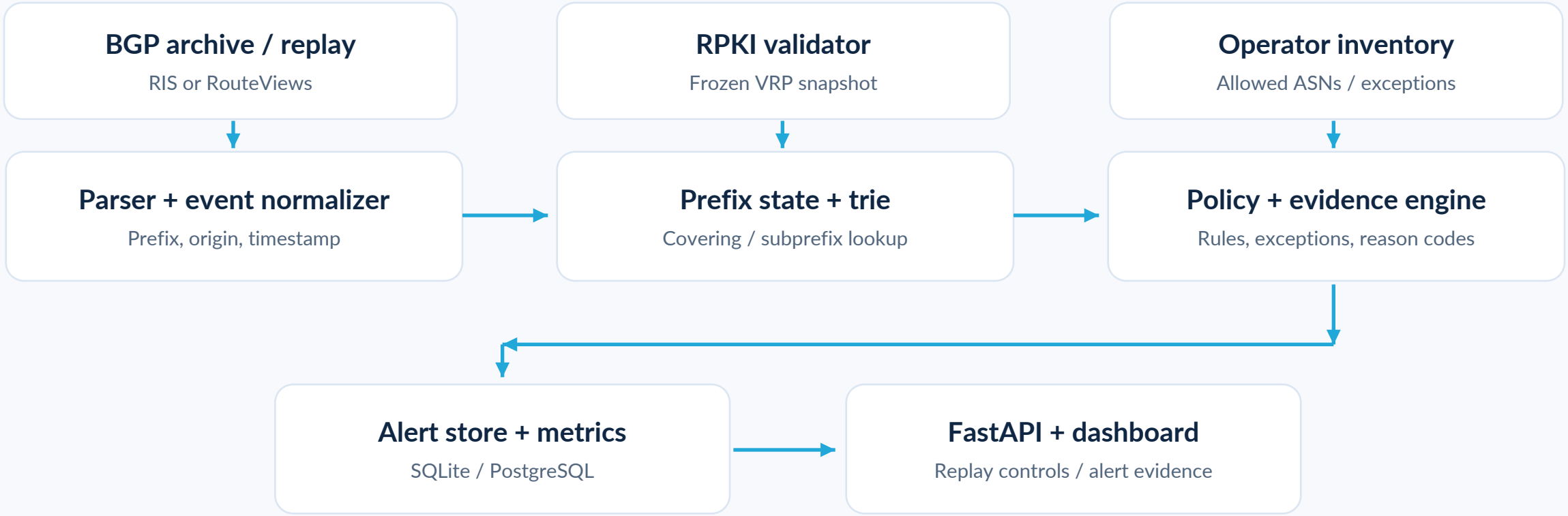
Proposed optimization vs. a simple baseline

Dimension	Baseline monitor	OriginGuard-X proposal
Detection context	ROA state only	**ROA + inventory + prefix history**
Subprefix analysis	Limited / manual	**Explicit covering-prefix lookup**
Legitimate changes	No formal exceptions	**MOAS / delegation / migration controls**
Explanation	Generic alert label	**Reason codes + matched evidence**
Evaluation	Aggregate detection count	**Precision, recall, FPR, P95 latency**

Ablation: ROA only / inventory only / combined / combined + subprefix + history

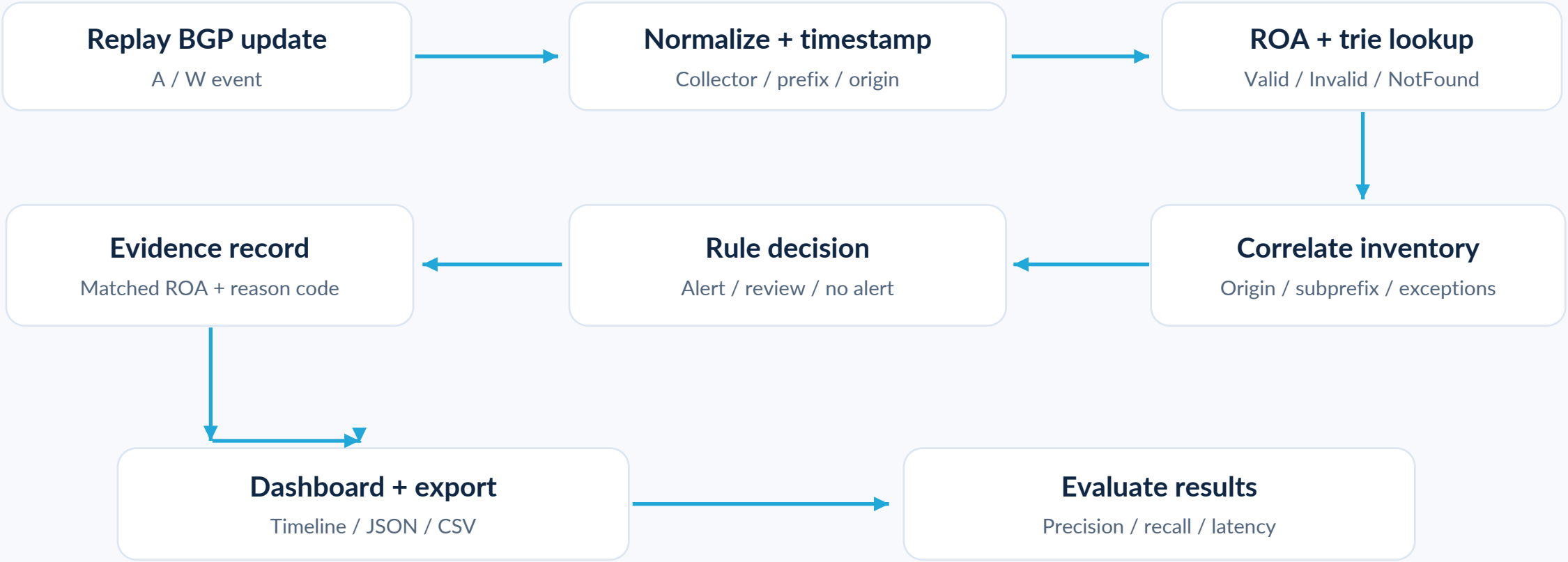
Solution architecture & design

Replay-based, read-only security monitoring architecture.



Data flow and decision logic

Every alert must be reproducible from event, snapshot, inventory and rule version.



No route changes, no automated mitigation, no full AS_PATH analysis.

OriginGuard-X: project deployment timeline

Proposed engineering milestones and deliverables - not the organizers' event agenda.

01 — 14 OCT

Scope + evidence design

Freeze baseline rules, threat cases, alert schema; submit the proposal.

02 — 16-20 OCT

Replay pipeline + snapshots

Parse BGP MRT; pin ROA/VRP and operator inventory snapshots.

03 — 21-25 OCT

Origin + subprefix detection

Implement RPKI states, prefix trie, and legitimate-change exceptions.

04 — 26-31 OCT

Explanations + demo interface

Generate reason-coded evidence; add API and replay dashboard.

05 — 1-4 NOV

Controlled evaluation

Test labelled attacks and benign cases; measure precision, recall, FPR, latency.

06 — 5-23 NOV*

Deployment + quality checks

Package read-only replay demo; run reproducibility and smoke tests.

*** Later-stage deployment is conditional on qualification.**
All demos are read-only; no automated route blocking or mitigation.

Selected literature & RFC references

Official standards and verified publisher records; not presented as a completed 10-paper review.

[1] **Mohapatra et al.**
RFC 6811: BGP Prefix Origin Validation (2013).
doi.org/10.17487/RFC6811

[2] **Snijders et al.**
RFC 9582: A Profile for ROAs (2024).
doi.org/10.17487/RFC9582

[3] **Bush**
RFC 8481: Clarifications to BGP Origin Validation (2018).
doi.org/10.17487/RFC8481

[4] **Sermpezis et al.**
ARTEMIS: Neutralizing BGP Hijacking Within a Minute.
IEEE/ACM ToN 26(6), 2018. doi:10.1109/TNET.2018.2869798

[5] **Orsini et al.**
BGPStream: A Software Framework for Live and Historical BGP Data Analysis.
ACM IMC 2016. doi:10.1145/2987443.2987482

[6] **RIPE NCC**
Routing Information Service (RIS).
https://ris.ripe.net/

[7] **University of Oregon**
RouteViews Project.
https://www.routeviews.org/

THANK YOU

Questions & discussion

Cipher Strike | A3-PS005-TC048

OriginGuard-X - Explainable BGP Anomaly Monitoring

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